Chapter 2: Moments

Part I — Statistics Foundation for ML

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Abstract

A distribution is an infinite-dimensional object; moments are the finite, computable summaries we actually work with – the mean, the variance, the covariance between two variables, and, packaged together, the moment generating function. Every loss function you will minimize later in this course (mean squared error, the Gaussian log-likelihood, the bias–variance decomposition) is a statement about moments. This chapter builds them carefully, in scalar and then matrix form, because the matrix form is exactly the shape you will need for multivariate regression and for the covariance matrices behind PCA (Part II, Chapter 5).

Acknowledgement: this chapter expands the STAT4ML lecture slides ("Slide 2 – Moments"), following Casella & Berger, Statistical Inference*, 2nd ed. (2002).*

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1 Moments

Definition 1.1. For integer n, the *nth moment* of X is $\mu'_n = E[X^n]$. The *nth central moment* of X is $\mu_n = E[(X-\mu)^n]$, where $\mu = \mu'_1 = E[X]$.

1.1 Expected Value

An everyday "average" weights every observed data point equally. The *expected value* instead weights every *possible* value a random variable can take by how probable that value is. It answers: on average, which value do we expect X to take?

Definition 1.2. The expected value of $g(X)$ is

$$E[g(X)] = \begin{cases} \displaystyle\int_{-\infty}^{\infty} g(x)\, f_X(x)\, dx & X \text{ continuous} \\ \displaystyle\sum_{x \in \mathcal{X}} g(x)\, f_X(x) & X \text{ discrete,} \end{cases}$$

provided the sum/integral converges absolutely; if $E[|g(X)|] = \infty$ we say $E[g(X)]$ does not exist.

Example 1.1 (Exponential mean)**.** Let $X \sim \text{Exponential}(\lambda)$, $f_X(x) = \frac{1}{\lambda} e^{-x/\lambda}$ for $x \geq 0$. Integrating by parts ($\int u\, dv = uv - \int v\, du$, with $u = x$, $dv = \frac{1}{\lambda} e^{-x/\lambda} dx$),

$$E[X] = \int_0^{\infty} x \cdot \frac{1}{\lambda} e^{-x/\lambda}\, dx = \Big[-x e^{-x/\lambda} \Big]_0^{\infty} + \int_0^{\infty} e^{-x/\lambda}\, dx = \lambda.$$

Expectation is a *linear* operator: for constants a, b,

$$E[a + bX] = a + b\, E[X] = a + b\mu.$$

Theorem 1.1. Let a, b, c be constants and g_1, g_2 functions whose expectations exist.

1. $E[a\, g_1(X) + b\, g_2(X) + c] = a\, E[g_1(X)] + b\, E[g_2(X)] + c$.
2. If $g_1(x) \geq 0$ for all x, then $E[g_1(X)] \geq 0$.
3. If $g_1(x) \geq g_2(x)$ for all x, then $E[g_1(X)] \geq E[g_2(X)]$.
4. If $a \leq g_1(x) \leq b$ for all x, then $a \leq E[g_1(X)] \leq b$.

Example 1.2. Let $X \sim \text{Uniform}(0, 1)$ and $g(X) = -\log X$. Then, using integration by parts,

$$E[-\log X] = \int_0^1 -\log x\, dx = \Big[-x \log x + x \Big]_0^1 = 1.$$

1.2 Variance

Variance measures the spread of X around its mean. While the mean is usually written μ, variance is written σ^2; it is a second-order moment.

Definition 1.3. The variance of X is its second central moment,

$$\text{Var}(X) = E[(X - \mu)^2], \qquad \sqrt{\text{Var}(X)} = \text{standard deviation of } X.$$

Equivalently, expanding the square and using linearity of expectation,

$$\text{Var}(X) = E[X^2] - \mu^2.$$

This second form, $E[X^2] - (E[X])^2$, is the one you will actually compute in code – it needs only two running sums (of x and of x^2) instead of a second pass over the data. It is also the single most common source of bugs in hand-written variance code: forgetting the $-\mu^2$ term always returns a number too large, and never negative, which makes the bug easy to miss on a quick sanity check. (This exact bug is planted in `exercise02_buggy.py` – watch for it.)

2 Covariance and Correlation

Definition 2.1. For random variables X, Y with $E[X] = \mu_X$, $E[Y] = \mu_Y$, $\mathrm{Var}(X) = \sigma_X^2$, $\mathrm{Var}(Y) = \sigma_Y^2$, the *covariance* is

$$\mathrm{Cov}(X, Y) = E\big[(X - \mu_X)(Y - \mu_Y)\big],$$

and the *correlation* (correlation coefficient) is

$$\rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

If large values of X tend to co-occur with large values of Y (and small with small), $\mathrm{Cov}(X, Y) > 0$: when $X > \mu_X$, $Y > \mu_Y$ is also likely, so $(X - \mu_X)(Y - \mu_Y) > 0$ on average. Symmetrically, if large values of X tend to pair with *small* values of Y, $\mathrm{Cov}(X, Y) < 0$. Correlation just rescales covariance by both standard deviations so that it is unitless and bounded in $[-1, 1]$ – which is what makes it comparable across different pairs of variables.

Theorem 2.1. $\mathrm{Cov}(X, Y) = E[XY] - \mu_X \mu_Y$.

Theorem 2.2. If $X \perp\!\!\!\perp Y$ (independent), then $\mathrm{Cov}(X, Y) = \rho_{XY} = 0$. The converse is **false** in general: zero correlation does not imply independence (a standard counterexample: $X \sim \mathrm{Uniform}(-1, 1)$, $Y = X^2$; then $\mathrm{Cov}(X, Y) = 0$ but Y is a deterministic function of X).

Theorem 2.3. For random variables X, Y and constants a, b,

$$\mathrm{Var}(aX + bY) = a^2 \mathrm{Var}(X) + b^2 \mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X, Y),$$

which reduces to $a^2 \mathrm{Var}(X) + b^2 \mathrm{Var}(Y)$ when X, Y are independent (since $\mathrm{Cov}(X, Y) = 0$).

If X, Y are positively correlated, $\mathrm{Var}(X + Y) > \mathrm{Var}(X) + \mathrm{Var}(Y)$ (the variances reinforce each other); if negatively correlated, $\mathrm{Var}(X + Y) < \mathrm{Var}(X) + \mathrm{Var}(Y)$ (they partially cancel).

2.1 Variance of Sums of Random Variables

For constants a_i and random variables X_i, $i = 1, \ldots, n$,

$$\mathrm{Var}\Big(\sum_{i=1}^{n} a_i X_i\Big) = \sum_{i=1}^{n} a_i^2 \mathrm{Var}(X_i) + \sum_{i \neq j} a_i a_j \mathrm{Cov}(X_i, X_j).$$

2.2 Higher Moments

The third and fourth (central, standardized) moments – skewness and kurtosis – describe how asymmetric and how fat-tailed a distribution is, respectively. They matter in ML whenever a model's error distribution is assumed Gaussian but the real residuals are heavy-tailed (e.g. financial returns, or gradient noise in SGD with a small batch size).

3 Moment Generating Functions

Definition 3.1. The *moment generating function* (MGF) of X is

$$M_X(t) = E[e^{tX}],$$

provided this expectation is finite for t in some open interval containing 0 (i.e. there is $h > 0$ such that $E[e^{tX}]$ exists for all $-h < t < h$). If no such neighborhood exists, the MGF does not exist.

Theorem 3.1. If X has MGF $M_X(t)$, then $E[X^n] = M_X^{(n)}(0)$, the nth derivative of M_X evaluated at $t = 0$. This is exactly what "moment generating" means: differentiate the MGF n times at zero and you get the nth raw moment, all at once.

3.1 The Normal MGF

For $X \sim N(\mu, \sigma^2)$ with $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$, completing the square in the exponent of $E[e^{tX}]$ gives

$$M_X(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right).$$

Differentiating,

$$E[X] = M_X'(0) = \mu, \qquad E[X^2] = M_X''(0) = \sigma^2 + \mu^2, \qquad \mathrm{Var}(X) = E[X^2] - (E[X])^2 = \sigma^2.$$

This is a good closed-form to numerically double-check: Exercise 2 asks you to estimate the Normal MGF by direct simulation (average e^{tX} over many samples) and compare against the formula above.

4 Matrix Notation for Moments

Let X be $(r \times 1)$ and Y be $(c \times 1)$ random vectors. The *cross-covariance matrix* is

$$\mathrm{Cov}(X, Y) = \begin{pmatrix} \mathrm{Cov}(X_1, Y_1) & \cdots & \mathrm{Cov}(X_1, Y_c) \\ \vdots & \ddots & \vdots \\ \mathrm{Cov}(X_r, Y_1) & \cdots & \mathrm{Cov}(X_r, Y_c) \end{pmatrix} = E\big[(X - E[X])(Y - E[Y])^\top\big].$$

For a single $(c \times 1)$ vector X, write $\mathrm{Cov}(X)$ for $\mathrm{Cov}(X, X)$:

$$\mathrm{Cov}(X) = \begin{pmatrix} \mathrm{Var}(X_1) & \cdots & \mathrm{Cov}(X_1, X_c) \\ \vdots & \ddots & \vdots \\ \mathrm{Cov}(X_1, X_c) & \cdots & \mathrm{Var}(X_c) \end{pmatrix},$$

a symmetric $(c \times c)$ matrix – *the* object PCA diagonalizes in Part II, Chapter 5.

For a block vector $X = \binom{Y}{Z}$ with Y $(p \times 1)$ and Z $(q \times 1)$,

$$\mathrm{Cov}(X) = \begin{pmatrix} \mathrm{Cov}(Y) & \mathrm{Cov}(Y, Z) \\ \mathrm{Cov}(Z, Y) & \mathrm{Cov}(Z) \end{pmatrix}.$$

For non-stochastic vectors a $(r \times 1)$, b $(c \times 1)$, and $\Sigma_{XY} = \mathrm{Cov}(X, Y)$,

$$\mathrm{Cov}(a^\top X, b^\top Y) = a^\top \Sigma_{XY} b.$$

In particular, with $\Sigma = \mathrm{Var}(X)$,

$$\mathrm{Var}(a^\top X) = a^\top \Sigma a,$$

and for two random vectors of the same dimension,

$$\mathrm{Var}(X+Y) = \Sigma_{XX}+\Sigma_{XY}+\Sigma_{YX}+\Sigma_{YY}, \qquad \mathrm{Var}(a^\top(X+Y)) = a^\top\Sigma_{XX}a+2a^\top\Sigma_{XY}a+a^\top\Sigma_{YY}a$$

(using $a^\top\Sigma_{XY}a = a^\top\Sigma_{YX}a$, since both equal the scalar $\mathrm{Cov}(a^\top X, a^\top Y)$).

Finally, these results extend from vectors to matrices: for a non-stochastic matrix R,

$$E[RX] = R\,E[X], \qquad \mathrm{Var}(RX) = R\,\mathrm{Var}(X)\,R^\top = R\,\Sigma_{XX}\,R^\top.$$

This is the identity that tells you how a linear layer $z = Wx$ transforms the covariance of its input – the starting point for reasoning about activation statistics in a deep network.

5 Summary

Summary

- Moments ($E[X^n]$, central moments $E[(X-\mu)^n]$) are finite numerical summaries of a distribution.
- Variance and covariance are the second-moment objects that drive the bias–variance tradeoff (Part II, Chapter 2) and PCA (Part II, Chapter 5).
- The MGF packages every moment into one function via $E[X^n] = M_X^{(n)}(0)$.
- In matrix form, $\mathrm{Var}(RX) = R\Sigma R^\top$ is the identity you will use repeatedly once random vectors, not just random scalars, are on the table.

6 Exercises

See `exercises/python/exercise02.py`, `exercises/r/exercise02.R`, and `exercises/rust/exercise02.rs` (expected value, variance, covariance, correlation, and a Monte-Carlo check of the Normal MGF), plus the Rustlings-style broken counterparts in `exercise02_buggy.{py,R,rs}`.

Reference

Casella, G., & Berger, R. (2002). *Statistical Inference* (2nd ed.). Cengage Learning.